

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12398701
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Rayne; Damian

Convection-driven power generator

Abstract

A convection-driven power generator comprising a flow intake configured to supply fluid to the generator, a flow duct having a duct inlet and a duct outlet wherein the duct outlet is spaced downstream from the duct inlet along the flow duct, the duct inlet being fluidly coupled to the flow intake. A heating chamber fluidly is coupled to the duct outlet so as to receive fluid from the duct outlet, the heating chamber comprising an external wall configured to transmit light radiation incident thereon such that fluid within the heating chamber is heated by the transmitted light radiation. A flow exhaust is fluidly coupled to the heating chamber and configured to exhaust fluid heated by the heating chamber from the heating chamber. A turbine is arranged within the flow duct, downstream of the flow intake, and exposed to fluid flow through the flow duct such that when fluid flows through the flow duct the turbine is caused to rotate by the fluid flow; and at least one lens element is configured to focus the light radiation transmitted by the external wall within the heating chamber.

Inventors:	Rayne; Damian (London, GB)
Applicant:	Rayne; Damian (London, GB)
Family ID:	1000008777637
Appl. No.:	17/921403
Filed (or PCT Filed):	April 27, 2021
PCT No.:	PCT/EP2021/060961
PCT Pub. No.:	WO2021/219623
PCT Pub. Date:	November 04, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20230132257 A1	Apr. 27, 2023

Foreign Application Priority Data

GB

2006297

Apr. 29, 2020

Publication Classification

Int. Cl.: **F03G6/04** (20060101); **F03D9/00** (20160101); **F03D9/37** (20160101); **F03G6/06** (20060101)

U.S. Cl.:

CPC **F03G6/045** (20130101); **F03D9/007** (20130101); **F03D9/37** (20160501); **F03G6/04** (20130101); **F03G6/062** (20210801); F05B2240/131 (20130101); F05B2240/133 (20130101); F05B2260/24 (20130101)

Field of Classification Search

CPC: F03D (9/007); F03D (9/35); F03D (9/37); F03G (6/04); F03G (6/045); F03G (6/06); F03G (6/062); F03G (6/063); F05B (2240/131); F05B (2240/133); F05B (2260/24); F24S (2080/503)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3970409	12/1975	Luchuk	N/A	N/A
4016725	12/1976	Fiss	261/DIG.11	F01K 13/00
4224528	12/1979	Argo	290/55	F03D 15/10
4355628	12/1981	Watts	126/609	F24S 23/71
6772593	12/2003	Dunn	60/641.11	F03D 9/007
2004/0206086	12/2003	Kim	N/A	N/A
2005/0169742	12/2004	Kane	N/A	N/A
2007/0225143	12/2006	Kim et al.	N/A	N/A
2008/0156317	12/2007	Yangpichit	N/A	N/A
2009/0212570	12/2008	Le et al.	N/A	N/A
2010/0003636	12/2009	Wagner	N/A	N/A
2010/0034636	12/2009	Tsou	N/A	N/A
2011/0173980	12/2010	Yangpichit	N/A	N/A
2012/0321454	12/2011	Yu	415/191	F03D 1/04
2019/0186473	12/2018	Al-Sulaiman	N/A	F03G 6/045
2019/0249749	12/2018	Maiullo	N/A	N/A
2021/0091634	12/2020	Nash	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
106321361	12/2016	CN	N/A
102006024341	12/2006	DE	F03D 1/04
1423607	12/2005	EP	F03D 1/04
2471710	12/2010	GB	N/A
102188888	12/2019	KR	N/A
9711272	12/1996	WO	N/A
2006098662	12/2005	WO	N/A
WO-2011120531	12/2010	WO	F03G 6/045
WO-2018046979	12/2017	WO	F03G 6/045

OTHER PUBLICATIONS

International Search Report and Written Opinion for International Patent Application No. PCT/EP2021/060961, dated Sep. 22, 2021, 19 pages. cited by applicant

Primary Examiner: Laurenzi; Mark A

Assistant Examiner: Hu; Xiaoting

Attorney, Agent or Firm: Dorsey & Whitney LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is the national stage application of International Patent Application No. PCT/EP2021/060961 filed on Apr. 27, 2021 and entitled “A Convection-Driven Power Generator”, which claims priority to Great Britain Patent Application No. 2006297.2, filed on Apr. 29, 2020 and entitled “A Convection-Driven Power Generator”, the entire contents of which are hereby incorporated by reference herein in their entireties.

(2) The present invention relates to a convection-driven power generator and to a turbine.

(3) Existing power generators are heavy, difficult to transport and are complicated, requiring regular maintenance with many moving parts. Furthermore, many use fossil fuels and so are not particularly environmentally friendly as their operation causes harmful greenhouse gasses to be released into the environment. This also limits the environments in which they can be used or else measures must be taken for fume extraction.

(4) There therefore exists a need for an alternative power generator which is simple in design and operation, low weight, robust, easy to transport and assembly, produces a very low level of noise, and, in embodiments which generate power from sunlight, is less sensitive to the height of the sun in the sky and so is capable of providing power during extended lengths of time Such a power generator lends itself particularly to being used in remote locations or in developing countries.

(5) The present invention aims to alleviate, at least to a certain extent, the problems and/or address at least to a certain extent the difficulties associated with the prior art.

(6) According to a first aspect of the present invention, there is provided a convection-driven power generator comprising: a flow intake configured to supply fluid to the generator; a flow duct having a duct inlet and a duct outlet wherein the duct outlet is spaced downstream from the duct inlet along the flow duct, the duct inlet being fluidly coupled to the flow intake; a heating chamber fluidly coupled to the duct outlet so as to receive fluid from the duct outlet, the heating chamber comprising an external wall configured to transmit light radiation incident thereon such that fluid

within the heating chamber is heated by the transmitted light radiation; a flow exhaust fluidly coupled to the heating chamber and configured to exhaust fluid heated by the heating chamber from the heating chamber; a turbine arranged within the flow duct, downstream of the flow intake, and exposed to fluid flow through the flow duct such that when fluid flows through the flow duct the turbine is caused to rotate by the fluid flow; and at least one lens element configured to focus the light radiation transmitted by the external wall within the heating chamber.

(7) Thus, the flow intake may optionally be configured to supply fluid to the flow duct, for example from the external environment (i.e. from the fluid environment surrounding the generator, e.g. the surrounding air). Thus, the flow duct may optionally be configured to receive fluid from the external environment via the flow intake. The term external wall is meant as a wall of the heating chamber (that is, forming a part of the heating chamber itself, and therefore a part of the generator itself) which defines a fluid boundary between the generator and the external environment: thus the external/ambient fluid environment is on one side of the wall (the external side of the external wall). Optionally, the turbine is upstream of the duct outlet.

(8) Optionally, the convection-driven power generator is configured such that, when fluid is heated within the heating chamber by the transmitted light radiation, the heated fluid is caused to flow through the flow exhaust via convection (e.g. predominately or solely by convection) and thereby establishes a fluid current within the flow duct which causes fluid to enter the duct inlet and in turn to enter the generator through the flow intake. Thus, the transmitted light radiations established a fluid current or flow down the fluid duct.

(9) Optionally, the lens element is configured to substantially focus the transmitted light radiation to an area, locus or point which is within the heating chamber but spaced away from all interior surfaces thereof such that the focused light radiation directly heats the fluid with the heating chamber, e.g. fluid within an internal fluid heating space of the heating chamber. Optionally, the lens element is configured to substantially focus the transmitted light radiation to an area, locus or point which is within the heating chamber but spaced away from all light-receiving surfaces thereof such that the focused light radiation directly heats the fluid with the heating chamber, e.g. fluid within an internal fluid heating space of the heating chamber. Optionally, the lens element is configured to substantially focus the transmitted light radiation to an area, locus or point which is within the internal fluid heating space of the heating chamber but spaced away from all light-receiving surfaces thereof such that the focused light radiation directly heats the fluid with the heating chamber, e.g. fluid within the internal fluid heating space of the heating chamber. Thus, the primary method of heat transfer from the light radiation to the fluid is not by conduction, e.g. from a light-absorbing surface, but instead the light is focused to a concentrated point or area to form a point or area of increased temperature which is spaced away from all light-receiving surfaces or other such internal surfaces of the heating chamber. This provides increased efficiency of heat transfer to the fluid as less heat is absorbed by any light-absorbing surfaces which may otherwise then be conducted away from the light-receiving surfaces into other parts or components of the generator and not directly to the fluid itself. Thus a more efficient generator is provided.

Furthermore, directly heating the fluid itself is particularly advantageous as only a predetermined and relatively small area of fluid within the heating chamber is heated and so the surrounding fluid remains cooler and may act as a thermal insulation between the heated air and the surrounding components or surfaces of the convection-driven generator, reducing thermal losses, and therefore improving the thermal efficiency of the convection-driven generator.

(10) Optionally, the lens element has a focal point or locus which is within the heating chamber but spaced away from all interior surfaces thereof such that the focused light radiation directly heats the fluid with the heating chamber, e.g. fluid within an internal fluid heating space of the heating chamber. Optionally, the lens element has a focal point or locus which is within the heating chamber but spaced away from all light-receiving surfaces thereof such that the focused light radiation directly heats the fluid with the heating chamber, e.g. fluid within an internal fluid heating

space of the heating chamber. This provides increased efficiency of heat transfer to the fluid as less heat is absorbed by any light-receiving surfaces which may then be conducted away from the fluid into other parts or components of the generator and not directly to the fluid itself. Thus a more efficient generator is provided.

(11) Optionally, the external wall of the heating chamber comprises the at least one lens element. Optionally, the external wall may be shaped so as to form the lens element, for example it may comprise one or more convex surfaces configured to act as a lens element, for example two back-to-back convex surfaces. Thus, the need for a separate lens element is avoided and the generator may be simpler in design and require fewer components.

(12) Optionally, the convection-driven power generator further comprises a chimney, the chimney comprising the flow duct and, optionally, the chimney also comprising the flow intake. Optionally, the flow duct passes through the chimney, e.g. longitudinally through the chimney, i.e. down the chimney. Optionally, the chimney surrounds at least a portion of the flow duct, i.e. the chimney forms a flow duct housing. The chimney is generally or substantially vertically aligned in use and may also be referred to as a tower. Optionally, the flow intake is provided at the highest end of the chimney, that is the opposite end of the chimney to the duct outlet, when the chimney comprises the duct outlet. Optionally, the flow duct may be provided within the chimney.

(13) Optionally, the heating chamber comprises an annulus extending therethrough, and wherein the flow duct, or a portion thereof, passes longitudinally through the annulus. Optionally the flow duct passes through generally the centre of the heating chamber, e.g. to thereby define a toroidal fluid heating space within the heating chamber, or to thereby define an annulus through which the flow duct passes longitudinally. Such an arrangement provides a particularly compact generator.

(14) Optionally, the heating chamber comprises an annulus extending therethrough, and wherein the chimney, or a portion thereof, passes longitudinally through the annulus. Such an arrangement provides a particularly compact generator. Optionally, the annulus is generally at the centre of the heating chamber. Optionally, the flow duct outlet is arranged towards or at the bottom/base of the heating chamber, e.g. towards the bottom of the internal fluid heating space such that fluid received from the fluid duct enters the heating chamber or internal fluid heating space towards or at its bottom/base.

(15) Optionally, the heating chamber is substantially or generally toroidal, wherein the annulus of the toroid is arranged substantially along the toroid axis of the heating chamber such that the heating chamber extends circumferentially around the flow duct. Such an arrangement provides a particularly compact generator. Optionally, the heating chamber is substantially or generally oblate toroidal.

(16) Optionally, the annulus and/or flow duct and/or chimney is arranged in use such that its longitudinal axis is substantially or generally vertical. Such an arrangement provides a particularly compact generator.

(17) Optionally, the flow exhaust extends peripherally around the chimney or flow duct.

(18) Optionally, the flow exhaust is arranged in use above the heating chamber such that heated fluid is caused to exit the flow exhaust by convection (solely or predominately by convection).

(19) Optionally, the external wall is made from plastic sheeting supported by a support structure, for example ribs, e.g. longitudinally and/or laterally extending ribs. Such a solution provides a lightweight means of supporting the plastic sheeting.

(20) Optionally, the external wall comprises two light transmitting layers spaced from each other, for example by an air gap, configured such that incident light passes through both of the two layers. The gap between the two layers provides a lighter external wall as well as functioning as a thermal insulating layer between the two layers, thereby reducing thermal losses and improving the efficiency of the generator.

(21) Optionally, the two layers of the external wall are arranged with respect to each other so as to form the lens element. Thus, the external wall may comprise or form the lens element, reducing the

number and complexity of parts required.

(22) Optionally, the flow duct comprises a convergent-divergent nozzle for accelerating the fluid flow, the turbine being provided within the nozzle, for example at the throat of the nozzle. The nozzle therefore accelerates the fluid flow such that it is fast enough to rotate the turbine to above the turbine stall speed. Such a configuration enables the turbine to rotate even when the incident light is so weak as to only generate a weak flow through the flow duct.

(23) Optionally, the flow intake comprises a variable flow restrictor configured to vary the fluid flow rate flowable through the generator. Optionally, the flow intake comprises a variable flow restrictor configured to vary the fluid flow rate flowable through the fluid duct. Thus, the flow restrictor may be configured to vary the power output of the generator by varying the fluid flow rate through the flow duct and thereby through the generator.

(24) Optionally, the variable flow restrictor is configured to vary the cross-sectional area of the flow intake so as to control the fluid flow rate flowable through the generator.

(25) Optionally, the variable flow restrictor comprises a collar or plug which is axially moveable with respect to the flow duct or chimney so as to vary the cross-sectional area of the flow intake and thereby control the fluid flow rate flowable through the generator.

(26) Optionally, the flow exhaust comprises a variable flow restrictor configured to vary the cross-sectional area of the flow exhaust so as to control the rate of fluid flow flowable through the generator. Optionally, the flow intake comprises a variable flow restrictor configured to vary the fluid flow rate flowable through the fluid duct. Thus, the flow restrictor may be configured to vary the power output of the generator by varying the fluid flow rate through the flow duct and thereby through the generator.

(27) Optionally, the flow exhaust variable flow restrictor is pivotably mounted to the generator, for example to the chimney or heating chamber, such that pivoting the flow restrictor varies the cross-sectional area of the flow exhaust. Optionally, the flow exhaust variable flow restrictor comprises a collar or plug which is axially moveable with respect to the flow duct or chimney so as to vary the cross-sectional area of the flow exhaust and thereby control the fluid flow rate flowable through the generator.

(28) Optionally, the fluid is air. Optionally the fluid is water.

(29) Optionally, the external wall of the heating chamber is configured to transmit solar radiation incident thereon such that fluid within the heating chamber is heated, e.g. directly heated, by the transmitted solar radiation.

(30) Optionally, the power generator is a light-electrical power generator configured to convert incident light radiation into electrical power.

(31) Optionally, the turbine is operably connected to an electrical generator such that rotation of the turbine causes the electrical generator to generate electrical power. Optionally, the turbine is operably connected to the electrical generator by a shaft.

(32) Optionally, the flow duct comprises stator blades arranged upstream of the turbine, the stator blades being configured to cause the flow to spiral down the flow duct before it passes through the axial turbine, for example the stator blades are configured to impart rotational momentum to the flow such that the flow is caused to rotate about the longitudinal axis of the flow duct as the fluid flows along the duct. The stator blades condition the air flow to impart rotation to the flow along the duct so as to more readily cause the turbine to rotate.

(33) Optionally, the turbine comprises a chamber containing a viscous liquid, wherein the chamber is configured such that the viscous liquid is caused to flow radially outwards within the chamber through centrifugal acceleration of the viscous liquid upon rotation of the turbine. Optionally the chamber extends between 5% and 95% of the radius of the turbine, for example between about 10% and 90%, about 15% and 85%, about 20% and 80%, about 30% and 70%, about 40% and 60%, between 50% and 100% or any combination thereof, for example between 10% and 70%, or between 50% and 60%.

- (34) Optionally, the turbine is supported by magnetic bearings spaced apart from each other along the rotational axis of the axial turbine. Magnetic bearings reduce frictional losses.
- (35) Optionally, the external wall is substantially or generally arcuate, e.g. substantially or generally convex. Being substantially arcuate means that the external wall may act as a lens element.
- (36) Optionally, the lens element comprises a substantially or generally arcuate portion of the external wall.
- (37) Optionally, a portion of the flow duct downstream of the turbine arcuately flares radially outwards to form a generally or substantially conical shape, the end of the flared portion comprising the duct outlet being fluidly coupled to the heating chamber.
- (38) Optionally, flow intake is configured to receive fluid from the external environment, for example ambient air.
- (39) Optionally, the flow exhaust is configured to exhaust the fluid to the external environment, for example to the ambient air.
- (40) Optionally, the turbine is an axial turbine.
- (41) Optionally, the power generator is generally or substantially spherical or spheroid.
- (42) Optionally, the flow intake is arranged in use vertically higher than the flow exhaust.
- (43) Optionally, the power generator is substantially or generally symmetrical about the central vertical axis of the power generator, for example about the longitudinal axis of the flow duct.
- (44) According to a second aspect of the present invention, there is provided a turbine having an axis of rotation, the turbine comprising an internal fluid chamber for containing fluid, wherein the internal fluid chamber is configured such that fluid within the internal fluid chamber is caused to centrifugally flow within the internal chamber radially outwards from the axis of rotation of the turbine when the turbine is rotating, thereby increasing the rotational inertia of the turbine.
- (45) Optionally, the internal fluid chamber is shaped such that as the rate of rotation of the turbine increases, fluid is caused to centrifugally flow further from the axis of rotation, such that increasing rates of rotation of the turbine increase the rotational inertia of the turbine.
- (46) Optionally the chamber extends between 5% and 95% of the radius of the turbine, for example between about 10% and 90%, about 15% and 85%, about 20% and 80%, about 30% and 70%, about 40% and 60%, between 50% and 100% or any combination thereof, for example between 10% and 70%, or between 50% and 60%.
- (47) Optionally, the internal fluid chamber comprises a wall having an arcuate surface which curves outwardly, along the axis of rotation of the turbine, with respect to the axis of rotation of the turbine along the axis of rotation. Thus, the surface is configured such that, when the turbine is not rotating, fluid within the internal fluid chamber is accumulated at bottom of the internal chamber and, as the turbine rotates, fluid is caused to flow radially outwards with respect to the rotational axis of the turbine to thereby increase the rotational inertia of the turbine. Optionally, the arcuate wall of the turbine is shaped such that, along the rotational axis of the turbine (i.e. from one end of the turbine towards the other), it becomes further away from the rotational axis of the turbine. Optionally, the arcuate surface forms the base of the internal fluid chamber. Optionally, the internal fluid chamber comprises a fluid well in which the fluid accumulates when the turbine is not rotating. Optionally, the fluid well is deepest towards the rotational axis of the turbine and becomes increasingly more shallow away from the rotational axis of the turbine.
- (48) Optionally, the internal fluid chamber is closed such that the fluid is unable to flow out of the internal fluid chamber.
- (49) Optionally, the internal fluid chamber contains fluid.
- (50) Optionally, the turbine is supported by magnetic bearings spaced apart from each other along the rotational axis of the turbine. Magnetic bearings provide reduced frictional losses.
- (51) Optionally, the turbine is an axial turbine.
-

Description

- (1) The present invention can be carried out in various ways and preferred embodiments of a convection-driven power generator and a turbine in accordance with the invention will now be described by way of example with reference to the accompanying drawings, in which:
- (2) FIG. 1 is a side view of a preferred embodiment of a convection-driven power generator according to a first embodiment of the present invention;
- (3) FIG. 2 is a side cross-sectional view of the embodiment of FIG. 1, taken through a central vertical plane of the power generator;
- (4) FIG. 3 is a side view of a preferred embodiment of a convection-driven power generator according to a second embodiment of the present invention, the second embodiment being substantially spheroidal in shape;
- (5) FIG. 4 is a top down view of the embodiment of FIG. 3;
- (6) FIG. 5 is a side cross-sectional view of the embodiment of FIG. 3, taken through a central vertical plane of the power generator;
- (7) FIG. 6 is a side cross-sectional schematic diagram of a flow duct according to certain principles of the present invention and as used in each of the embodiments disclosed herein;
- (8) FIG. 7 is a side partial cross-sectional view of a preferred embodiment of a convection-driven power generator according to a third embodiment of the present invention;
- (9) FIGS. 8, 9 and 10 are a top and bottom view and a side cross-sectional view respectively of a preferred embodiment of a turbine according to the present invention.
- (10) A convection-driven power generator 1 according to a first aspect of the present invention is shown in FIG. 1. The convection-driven power generator comprises a vertically arranged chimney 2 which extends out of the top of, and at the centre of, a heating chamber 4. The chimney 2 of this embodiment, and all other embodiments described herein, may also be referred to as a tower 2, such as a flow tower 2. The chimney 2 comprises a flow duct 3 passing longitudinally and substantially coaxially therethrough. The flow duct 3 is therefore substantially or generally vertically arranged within the power generator 1. The chimney 2, at least partially, houses the flow duct 3 and therefore can also be referred to as a flow duct housing 2. In other words, the flow duct 3 extends vertically out of the top of, and in the centre of, the heating chamber 4 and the chimney forms a flow-duct housing for housing the portion of the flow duct 3 which extends outside of the heating chamber 4. A flow intake 5 is provided at or towards a distal end of the chimney 2, namely the highest end of the chimney 2, and which is fluidly coupled to the flow duct 3. The flow intake 5 is configured to supply fluid to the flow duct 3. In particular, the flow intake 5 may in some embodiments be configured to supply ambient fluid to the chimney 2, for example ambient air and thus may serve as an ambient air intake for supplying ambient air to the flow duct 3. In the embodiment of FIG. 1 the flow intake 5 is configured as a slot which extends around the periphery of the chimney 2. A turbine 10 (FIG. 2) is arranged within the flow duct 3, specifically within the portion of the flow duct which is within the heating chamber 4 (i.e. not the portion of the flow duct 3 which extends vertically out of the top of the heating chamber 4).
- (11) The flow intake 5 comprises a variable flow-restrictor 18 configured to vary the flow rate through the flow intake 5 and thereby through the flow duct 3 by varying the resistance to the flow. As the variable flow-restrictor 18 varies the flow rate through the flow duct 3, it also varies the rotation-rate of the turbine 10 and thereby the power output of the generator 1. In this embodiment, this is achieved by varying the cross-sectional area of the flow intake 5 to thereby allow less or more fluid, e.g. air, into the flow duct 3. In this embodiment, the variable flow-restrictor 18 comprises a plug 19 which is axially moveable with respect to the flow duct 3 or chimney 2 so as to vary the cross-sectional area of the flow intake 5 and thereby control the fluid flow rate flowable through the generator 1. In other embodiments a collar 20 may alternatively or additionally be

provided which extends around the periphery of the chimney **2** and which is also axially moveable with respect to the flow duct **3** or chimney **2** so as to vary the cross-sectional area of the flow intake **5** and thereby control the fluid flow rate flowable through the generator **1**. In these embodiments, the flow intake **5** may comprise a slot **5** which extends peripherally around the chimney **2** and which is defined by the gap between the chimney **2** and the axially-moveable plug **19**.

(12) A heating chamber **4** is provided at the opposite end of the chimney **2**, and therefore flow duct **3**, to the flow intake **5**. The flow duct **3** passes vertically through the centre of the heating chamber **4** which, in this embodiment, is generally or substantially an oblate spheroid in shape although other configurations and shapes of the heating chamber **4** may alternatively be used as will be seen below. Thus, the heating chamber **4** may be said to be substantially or generally toroidal, or oblate toroidal, in shape as it defines a central aperture or annulus **12** through which the flow duct **3** passes and therefore defines a substantially toroidal fluid space within the heating chamber **4**.

(13) Spaced between the distal end of the chimney **2** and the heating chamber **4**, a flow exhaust **9** is provided. The flow exhaust **9** is configured to allow fluid to flow out of the heating chamber **4** by convection (in this embodiment, solely by convection, however example are envisaged where convection may be assisted, e.g. by a fan, and in such embodiments, the fluid may exist the flow exhaust predominantly by convection) and, in some embodiments where the fluid is air, to the ambient air surrounding the power generator **1**. Thus the flow exhaust **9** is configured to allow heated fluid to exit the power generator **1**.

(14) The flow exhaust **9** also comprises a variable flow-restrictor **21** configured to vary the flow rate through the flow exhaust **9** by varying the resistance to the flow through the flow exhaust **9**. The exhaust variable flow-restrictor **21** is in this embodiment configured to vary the cross-sectional area of the flow exhaust so as to control the rate of fluid flow flowable through the flow duct **3** and thereby through the generator **1**. In the embodiment of FIG. **1**, the exhaust variable flow-restrictor **21** comprises one or more panels **21** which are pivotably mounted to the generator **1**, for example to the chimney **2** or heating chamber **4**, such that pivoting the panels **21** causes the cross-sectional area of the flow exhaust **9** to be varied.

(15) It is not necessary for the power generator **1** to require both the intake variable flow-restrictor **18** as well as the exhaust flow-restrictor **21** in order to control fluid flow through the generator **1**—either one, or both, may optionally be provided.

(16) The proximal end of the flow duct **3**, that is the end of the flow duct **3** which is opposite to the end fluidly coupled to the flow intake **5**, comprises a flow duct outlet **7** which is fluidly coupled to the heating chamber **4** such that fluid from the flow duct **3** may flow into the heating chamber **4**. In this embodiment, although it is not essential, the proximal end of the flow duct **3** is configured to arcuately flare radially outwards to form a generally or substantially conical shape. The flow duct outlet **7** comprises a circumferentially extending slot which comprises a plurality of radially spaced strakes **27** configured to segment the flow and to condition it in order to reduce the turbulence of the flow and may optionally be configured to provide laminar flow to the heating chamber **4**.

(17) The heating chamber **4** comprises an external wall **8** which is configured to transmit light radiation incident thereon such that fluid within the heating chamber is heated by the transmitted light radiation. The heating chamber **4** therefore defines an internal fluid heating space **31** within which the fluid is heated by the light radiation transmitted by the external wall **8**. In this embodiment and in the others described herein, the light radiation may optionally be solar light radiation.

(18) In this embodiment, the external wall **8** is optionally made from plastic sheeting which is optionally supported by a support structure which may optionally comprise ribs such as longitudinally extending ribs **13** and/or laterally extending ribs **14**. Equally, glass or other such light transmitting materials may also be used for the external wall **8** instead of the plastic sheeting and which may not require a support structure.

(19) The external wall **8** by virtue of its arcuate shape is configured to substantially focus the light

radiation incident on it into the internal fluid heating space **31** within the heating chamber **4**. Thereby, the power-generator **1** increases the heating effect of the transmitted light and improves the efficiency of the generator **1**. Optionally, to further improve the heating effect and efficiency of the generator **1**, the generator **1**, for example the external wall **8**, may comprise a lens element **11** configured to focus the light radiation transmitted by the external wall **8** within the heating chamber. The lens element **11** may have a focal point or focal locus **32** which is within the heating chamber but is spaced away from all interior surfaces thereof (e.g. spaced away from all light receiving surfaces thereof) such that the focused light radiation directly heats the fluid within the heating chamber, i.e. the fluid within the internal fluid heating space **31** of the heating chamber. Thus, the fluid is heated solely, or at least predominantly, directly by the transmitted light. Optionally, the lens element **11** may be configured to focus the transmitted light radiation to an area, locus or point **32** which is within the heating chamber **4** but spaced away from all interior surfaces thereof (including the external wall **8**, for example all surfaces which define the internal fluid heating space) such that the focused light radiation directly heats the fluid within the heating chamber **4**. In the embodiment of FIG. **1**, the arcuate external wall **8** itself, by virtue of its arcuate shape, comprises the lens element **11**. Thus, the external wall **8** itself may be configured so as to focus the transmitted light radiation to an area, locus or point **32** which is within the heating chamber **8** but spaced away from all interior surfaces thereof such that the focused light radiation directly heats the fluid within the heating chamber **4**. The locus **32** may extend circumferentially around the flow duct **3** within the heating chamber **4** to form a heating zone which is substantially ring-shaped or toroidal.

(20) As in some embodiments the arcuate shape of the external wall **8** itself may form the lens element **11**, the external wall **8** may optionally comprise two spaced-apart layers **14**, **16** of material. The two layers of material may have an air-gap therebetween to reduce the thermal losses to the environment or the two layers may have a space therebetween which is configured to be filled with water, or another suitable liquid, thereby improving the focusing effect of the lens element **11**.

(21) As can be seen in FIG. **2**, which shows a cross sectional-view of the embodiment of FIG. **1**, the flow duct **3** comprises a convergent divergent nozzle **17**. The convergent divergent nozzle **17** is configured to accelerate the fluid flow within the flow duct **3**. An axial turbine **10** is provided at the throat of the nozzle **17**, such that it is exposed to the flow within the flow duct **3**. Providing the turbine **10** at the throat of the nozzle **17** means that it is provided with accelerated fluid flow as the nozzle **17** accelerates the flow from a speed which may not otherwise be optimal to drive turbine **10**. The turbine **10** is operably coupled to an electrical generator **22** such that rotation of the turbine causes the electrical generator **22** to generate electrical power. A shaft forms the operable connection between the turbine **10** and the electrical generator **22**.

(22) Although not shown in the drawings, the first embodiment, as well as all of the embodiments described herein, may comprise stator blades arranged upstream of the turbine. The stator blades are configured to cause the flow to spiral down the flow duct before it passes through the turbine. Thus, the stator blades are configured to impart rotational momentum to the flow such that the flow is caused to rotate about the longitudinal axis of the flow duct as the fluid flows along the duct. The stator blades therefore cause the turbine to rotate more readily as the fluid is already rotating.

(23) The operation of the convection-driven generator will now be described, with reference to FIG. **2**. As light radiation **28** falls on the external wall **8** of the heating chamber **4**, it is transmitted by the external wall **8** and focused by the lens element **11** into the interior fluid heating space **31** of the heating chamber **4** which is filled with fluid to be heated.

(24) The focused light may in some embodiments be focused a point or a locus **32** which is spaced away from the interior surfaces of the heating chamber **4** such that the transmitted and focused light directly heats the fluid within the heating chamber **4**. Thus the lens element **11** is configured to form a heating zone which is spaced away from the interior walls or surfaces of the heating chamber **4** such that the transmitted and focused light directly heats the fluid within the heating

chamber. Thus, the fluid within the heating chamber **4** in some embodiments may not be heated by conduction, or at least not predominately by conduction, e.g. by conduction of heat from an interior surface of the heating chamber **4**, but is instead heated predominately directly by the transmitted light within the heating zone.

(25) As the fluid within the heating chamber **4** is heated by the transmitted light radiation **28**, it begins to rise upwards by convection and escapes the heating chamber **4**, and the generator **1**, through the flow exhaust **9**. The temperature to which the fluid is heated is controlled by the exhaust variable flow restrictor **21** by controlling the rate at which fluid is able to flow out of the heating chamber **4**. Thus, the exhaust variable flow restrictor **21** is pivoted to a position in order to achieve the desired temperature within the heating chamber **4** and thereby the desired fluid flow rate out of the heating chamber **4**. Once the heated fluid passes through the exhaust variable flow restrictor **21**, it is then exhausted to the ambient surroundings.

(26) The escape of heated fluid from the heating chamber **4** causes a reduced pressure in the heating chamber **4** which causes more fluid to be drawn into the heating chamber **4** via flow duct outlet **7**, thereby pulling fluid down the flow duct **3** and establishing a fluid current through the flow duct **3**. This in turn causes un-heated, ambient air to enter the generator via the flow duct **3** via flow intake **5**, the flow rate of the un-heated fluid being controllable via the intake variable flow restrictor **18**.

(27) The established fluid current through the flow duct **3** drives rotation of the axial turbine **10** which thus generates electrical power via electrical generator **22** operably coupled to the turbine **10** via a shaft.

(28) As can be seen more readily in FIGS. **6** and **10**, the axial turbine **10** comprises an internal fluid chamber **23** containing, or configured to contain, a viscous liquid **26**. The internal fluid chamber **23** is configured such that the viscous liquid is caused to flow radially outwards within the fluid chamber through centrifugal acceleration of the viscous liquid upon rotation of the turbine. In particular, the internal fluid chamber **23** comprises a base having an arcuate internal surface which curves outwardly with respect to the axis of rotation of the turbine **10** along the axis of rotation, forming a flared-radially-outwardly dish, cone or fluid well having its deepest portion towards the centre of the axis of rotation of the turbine. As the turbine **10** rotates with increasing speed, so the viscous liquid **26** flow by centrifugal acceleration out of the well, radially outwards, thereby increasing the rotational inertia of the turbine. Thus, the performance of the turbine at higher fluid flow rates is improved, while reducing the chance of the turbine stalling at lower fluid flow rates. Thus, the range of operational rotation rates of the turbine is increased. The internal fluid chamber **23** may be closed, that is, it may contain the viscous fluid such that the viscous fluid may not flow out of the chamber **23**. Alternatively, examples are envisaged wherein the internal fluid chamber **23** has a fluid supply line for supplying/extracting fluid from the internal fluid chamber **23** so as to vary the quantity of viscous fluid within the internal fluid chamber **23** to thereby vary the rotational inertia as desired. A pump could be used for pumping the viscous fluid through the fluid supply line.

(29) The turbine **10** is supported by magnetic bearings **24** spaced apart from each other along the rotational axis of the turbine **10** and substantially concentric with the rotational axis of the turbine. The magnetic bearings **24** reduce frictional losses of the turbine, thereby improving the overall efficiency of the generator **1**, although other means of supporting the turbine **10** could alternatively be used. The internal fluid chamber **23** is arranged between the magnetic bearings **24**.

(30) Turning now to FIGS. **3** and **4**, a second embodiment of a convection-driven power generator is shown. Similar features as those of the embodiment of FIG. **1** are also shown but with the reference numbers increased by 100. The embodiment of FIG. **3** is substantially the same as that of FIG. **1**, having all of the same features of the embodiment of FIG. **1** except that the heating chamber is substantially spheroidal and extends higher up the flow duct than in FIG. **1**. The embodiment of FIG. **3** is preferred in operating environments in which the incident light shines

more in from the sides than it does down on top of the generator **1**, for example in locations towards the poles of the Earth or for use during winter.

(31) The convection-driven power generator **101** comprises a vertically arranged chimney **102** which extends within the centre of a substantially spherical heating chamber. The chimney **102** comprises a flow duct **103** passing longitudinally and coaxially therethrough in exactly the same way as in FIG. **1**. The flow duct **103** is therefore also substantially or generally vertically arranged within the power generator **101**. The chimney **102** houses the flow duct **103** and therefore can also be referred to as a flow duct housing **102**. A flow intake **105** is provided at or towards a distal end of the chimney **102**, namely the highest end of the chimney **102**, and which is fluidly coupled to the flow duct **103**. The flow intake **105** is configured to supply fluid to the flow duct **103**. In particular, the flow intake **105** may in some embodiments be configured to supply ambient fluid to the chimney **102**, for example ambient air and thus may serve as an ambient air intake for supplying ambient air to the flow duct **103**. In the embodiment of FIG. **1** the flow intake **105** is configured as a slot **105** which extends around the periphery of the chimney **102**.

(32) The flow intake **105** comprises a variable flow-restrictor **118** configured to vary the flow rate through the flow intake **105** by varying the flow rate through the flow duct **103** by varying the resistance to the flow. As the variable flow-restrictor **118** varies the flow rate through the flow duct **103**, it also varies the rotation-rate of the turbine **120** and thereby the power output of the generator **101**. In this embodiment, this is achieved by varying the cross-sectional area of the flow intake **105** to thereby allow less or more fluid, e.g. air, into the flow duct **103**. In this embodiment, the variable flow-restrictor **118** comprises a plug **119** which is axially moveable with respect to the flow duct **103** or chimney **102** so as to vary the cross-sectional area of the flow intake **105** and thereby control the fluid flow rate flowable through the generator **101**. Thus, the flow intake **105** may comprise a slot **105** which extends peripherally around the chimney **102** and which is defined by the gap between the chimney **102** and the axially-moveable plug **119**.

(33) A substantially spheroidal heating chamber **104** is provided such that the flow duct **103** passes vertically through the centre of the heating chamber **104**. Thus, the heating chamber **104** may be said to be substantially or generally toroidal in shape as it defines a central aperture or annulus **112** through which the flow duct **103** passes and therefore defines a substantially toroidal fluid space within the heating chamber **104**.

(34) Spaced between the distal end of the chimney **102** and the heating chamber **104**, a flow exhaust **109** is provided. The flow exhaust **109** is configured to allow fluid to flow out of the heating chamber **104** by convection and, in some embodiments where the fluid is air, to the ambient air surrounding the power generator **101**. Thus the flow exhaust **9** is configured to allow heated fluid to exit the power generator **101**.

(35) The flow exhaust **109** also comprises a variable flow-restrictor **121** configured to vary the flow rate through the flow exhaust **109** by varying the resistance to the flow through the flow exhaust **109**. The exhaust variable flow-restrictor **121** is in this embodiment configured to vary the cross-sectional area of the flow exhaust so as to control the rate of fluid flow flowable through the flow duct **103** and thereby through the generator **101**. In the embodiment of FIG. **3**, the exhaust variable flow-restrictor **121** comprises one or more panels **121** which are pivotably mounted to the generator **101**, for example to the chimney **102** or heating chamber **104**, such that pivoting the panels **121** causes the cross-sectional area of the flow exhaust **109** to be varied.

(36) It is not necessary for the power generator **101** to require both the intake variable flow-restrictor **118** as well as the exhaust flow-restrictor **121** in order to control fluid flow through the generator **101**—either one, or both, may optionally be provided.

(37) The proximal end of the flow duct **103**, that is the end of the flow duct **103** which is opposite to the end fluidly coupled to the flow intake **105**, comprises a flow duct outlet **107** which is fluidly coupled to the heating chamber **104** such that fluid from the flow duct **103** may flow into the heating chamber **104**. In this embodiment, although it is not essential, the proximal end of the flow

duct **103** is configured to arcuately flare radially outwards to form a generally or substantially conical shape.

(38) The heating chamber **104** comprises an external wall **108** which is configured to transmit light radiation incident thereon such that fluid within the heating chamber is heated by the transmitted light radiation. The heating chamber **104** therefore defines an internal fluid heating space **131** within which the fluid is heated by the light radiation transmitted by the external wall **108**. In this embodiment and in the others described herein, the light radiation may optionally be solar light radiation.

(39) In this embodiment, the external wall **108** is optionally made from plastic sheeting which is optionally supported by a support structure which may optionally comprise ribs such as longitudinally extending ribs **113** and/or laterally extending ribs **114**. Equally, glass or other such light transmitting materials may also be used for the external wall **108** instead of the plastic sheeting and which may not require a support structure.

(40) The external wall **108** by virtue of its arcuate shape is configured to substantially focus the light radiation incident on it into the internal fluid heating space **31** within the heating chamber **104**. Thereby, the power-generator **101** increases the heating effect of the transmitted light and improves the efficiency of the generator **101**. Optionally, to further improve the heating effect and efficiency of the generator **101**, the generator **101**, for example the external wall **108**, may comprise a lens element **111** configured to focus the light radiation transmitted by the external wall **108** within the heating chamber **104**. The lens element **111** may have a focal point or focal locus **132** which is within the heating chamber but is spaced away from all interior surfaces thereof (e.g. spaced away from all light receiving surfaces thereof) such that the focused light radiation directly heats the fluid within the heating chamber, i.e. the fluid within the internal fluid heating space **131** of the heating chamber. Thus, the fluid is heated solely, or at least predominantly, directly by the transmitted light. Optionally, the lens element **111** may be configured to focus the transmitted light radiation to an area, locus or point **132** which is within the heating chamber **104** but spaced away from all interior surfaces thereof (including the external wall **108**, for example all surfaces which define the fluid heating space) such that the focused light radiation directly heats the fluid within the heating chamber **104**. The embodiment of FIG. 3, as an alternative to the embodiment of FIG. 1, uses panes of glass which are shaped so as to form a lens element **111**, although the plastic sheeting arrangement of FIG. 1 could instead be used. The locus **132** may extend circumferentially around the flow duct **103** within the heating chamber **104** to form a heating zone which is substantially ring-shaped or toroidal.

(41) As in some embodiments the arcuate shape of the external wall **108** itself may form the lens element **11**, the external wall **8** may alternatively comprise two spaced-apart layers **114**, **116** of material. The two layers of material may have an air-gap therebetween to reduce the thermal losses to the environment or the two layers may have a space therebetween which is configured to be filled with water, or another suitable liquid, thereby improving the focusing effect of the lens element **111**.

(42) In the same way as in FIG. 2, which shows a cross sectional-view of the embodiment of FIG. 1, the flow duct **103** of the second embodiment of FIG. 3 comprises a convergent divergent nozzle **117**, although this is not explicitly shown in the figures. The convergent divergent nozzle **117** is configured to accelerate the fluid flow within the flow duct **103**. An axial turbine **110** is provided at the throat of the nozzle **117**, such that it is exposed to the flow within the flow duct **103**. Providing the turbine **110** at the throat of the nozzle **117** means that it is provided with accelerated fluid flow as the nozzle **117** accelerates the flow from a speed which may not otherwise be optimal to drive turbine **110**. The turbine **110** is operably coupled to an electrical generator **122** such that rotation of the turbine causes the electrical generator **122** to generate electrical power. A shaft forms the operable connection between the turbine **110** and the electrical generator **122**.

(43) The operation of the second embodiment of the convection-driven generator is substantially

the same as that of the first. With reference to FIG. 5, as light radiation falls on the external wall **108** of the heating chamber **104**, it is transmitted by the external wall **108** and focused by the lens element **111** into the interior fluid heating space **131** of the heating chamber **104** which is filled with fluid to be heated.

(44) The focused light may in some embodiments be focused to a point or a locus **132** which is spaced away from the interior surfaces of the heating chamber **104** such that the transmitted and focused light directly heats the fluid within the heating chamber **104**. Thus, the lens element **111** is configured to form a heating zone which is spaced away from all internal walls or surfaces of the heating chamber **104** such that the transmitted and focused light directly heats the fluid within the heating chamber **104**. Thus, the fluid within the heating chamber **104** in some embodiments may not be heated by conduction, or at least not predominantly by conduction, e.g. by conduction of heat from an interior surface of the heating chamber **104**, but is instead heated predominately directly by the transmitted light within the heating zone.

(45) As the fluid within the heating chamber **104** is heated by the transmitted light radiation **28**, it begins to rise upwards by convection and escapes the heating chamber **104**, and the generator **101**, through the flow exhaust **109**. The temperature to which the fluid is heated is controlled by the exhaust variable flow restrictor **121** by controlling the rate at which fluid is able to flow out of the heating chamber **104**. Thus, the exhaust variable flow restrictor **121** is positioned in order to achieve the desired temperature within the heating chamber **104** and thereby the desired fluid flow rate out of the heating chamber **104**. Once the heated fluid passes through the exhaust variable flow restrictor **121**, it is then exhausted to the ambient surroundings. In this embodiment, the flow restrictor **121** is slidable rather than pivotable and serves as a plug **121** to plug the flow exhaust **109**, thereby controlling the rate of fluid flow from the heating chamber **104** and through the generator **101**.

(46) The escape of heated fluid from the heating chamber **104** causes a reduced pressure in the heating chamber **104** which causes more fluid to be drawn into the heating chamber **104** via flow duct outlet **107**, thereby pulling fluid down the flow duct **103** and establishing a fluid current through the flow duct **103**. This in turn causes un-heated, ambient air to enter the generator **101** via the flow duct **103** via flow intake **105**, the flow rate of the un-heated fluid being controllable via the intake variable flow restrictor **118**.

(47) The established fluid current through the flow duct **103** drives rotation of the axial turbine **110** which thus generates electrical power via electrical generator **122** operably coupled to the turbine **110** via a shaft.

(48) As can be seen more readily in FIGS. 6 and 10, the axial turbine **110** comprises an internal chamber **23** containing, or configured to contain, a viscous liquid **26**. The internal chamber **23** is configured such that the viscous liquid is caused to flow radially outwards within the chamber through centrifugal acceleration of the viscous liquid upon rotation of the turbine. In particular, the internal chamber **23** comprises a base having an arcuate internal surface which curves outwardly with respect to the axis of rotation of the turbine **110** along the axis of rotation, forming a flared-radially-outwardly dish, cone or fluid well. As the turbine **110** rotates with increasing speed, so the viscous liquid **26** is pulled out of the fluid well **30** by centrifugal acceleration, thereby increasing the rotational inertia of the turbine. Thus, the performance of the turbine **110** at higher fluid flow rates is improved, while reducing the chance of the turbine **110** stalling at lower fluid flow rates.

(49) The turbine **110** is supported by magnetic bearings **24** spaced apart from each other along the rotational axis of the turbine **110** and substantially concentric with the rotational axis of the turbine **110**. The magnetic bearings **24** reduce frictional losses of the turbine, thereby improving the overall efficiency of the generator **101**, although other means of supporting the turbine **110** could alternatively be used.

(50) Turning now to FIG. 7, a third embodiment of a convection-driven power generator **201** is shown. Similar features as those of the embodiment of FIG. 1 are also shown but with the reference

numbers increased by 200. The embodiment of FIG. 7 is substantially the same as that of FIG. 1, having all of the same features of the embodiment of FIG. 1 except that the heating chamber comprises an external wall having a substantially tubular section 233 and a substantially conical section 234 arranged on top of the tubular section 233.

(51) The convection-driven power generator 201 of the third embodiment comprises a vertically arranged chimney 202 which extends out of the centre of a heating chamber 204. The chimney 202 of this embodiment, and all other embodiments described herein, may also be referred to as a tower 202, such as a flow tower 202. The chimney 202 comprises a flow duct 203 passing longitudinally and substantially coaxially therethrough. The flow duct 203 is therefore substantially or generally vertically arranged within the power generator 201. The chimney 202, at least partially, houses the flow duct 203 and therefore can also be referred to as a flow duct housing 202. A flow intake 205 is provided at or towards a distal end of the chimney 202, namely the highest end of the chimney 202, and which is fluidly coupled to the flow duct 203. The flow intake 205 is configured to supply fluid to the flow duct 203. In particular, the flow intake 205 may in some embodiments be configured to supply ambient fluid to the chimney 202, for example ambient air and thus may serve as an ambient air intake for supplying ambient air to the flow duct 203. In the embodiment of FIG. 1 the flow intake 205 is configured as a slot which extends around the periphery of the chimney 202.

(52) The flow intake 205 comprises a variable flow-restrictor 218 configured to vary the flow rate through the flow intake 205 and thereby vary the flow rate through the flow duct 203 by varying the resistance to the flow. As the variable flow-restrictor 218 varies the flow rate through the flow duct 203, it also varies the rotation-rate of the turbine 210 and thereby the power output of the generator 201. In this embodiment, this is achieved by varying the cross-sectional area of the flow intake 205 to thereby allow less or more fluid, e.g. air, into the flow duct 203. In this embodiment, the variable flow-restrictor 218 comprises a plug 219 which is axially moveable with respect to the flow duct 203 or chimney 202 so as to vary the cross-sectional area of the flow intake 205 and thereby control the fluid flow rate flowable through the generator 201. In other embodiments a collar 220 may alternatively or additionally be provided which extends around the periphery of the chimney 202 and which is also axially moveable with respect to the flow duct 203 or chimney 202 so as to vary the cross-sectional area of the flow intake 205 and thereby control the fluid flow rate flowable through the generator 201. In these embodiments, the flow intake 205 may comprise a slot 205 which extends peripherally around the chimney 202 and which is defined by the gap between the chimney 202 and the axially-moveable plug 219.

(53) A heating chamber 204 is provided at the opposite end of the chimney 202, and therefore flow duct 203, to the flow intake 205. The flow duct 203 passes vertically through the centre of the heating chamber 204 which, in this embodiment, comprises a substantially conical external wall 234 arranged on top of a substantially tubular external wall 233, although other configurations and shapes of the heating chamber 204 may alternatively be used. Thus, the heating chamber 204 may be said to be substantially or generally toroidal in shape as it defines a central passage, aperture or annulus 212 through which the flow duct 203 passes.

(54) Spaced between the distal end of the chimney 202 and the heating chamber 204, a flow exhaust 209 is provided. The flow exhaust 209 is configured to allow fluid to flow out of the heating chamber 204 by convection and, in some embodiments where the fluid is air, to the ambient air surrounding the power generator 201. Thus the flow exhaust 209 is configured to allow heated fluid to exit the power generator 201.

(55) The flow exhaust 209 also comprises a variable flow-restrictor 221 configured to vary the flow rate through the flow exhaust 209 by varying the resistance to the flow through the flow exhaust 209. The exhaust variable flow-restrictor 221 is in this embodiment configured to vary the cross-sectional area of the flow exhaust 209 so as to control the rate of fluid flow flowable through the flow duct 203 and thereby through the generator 201. In the embodiment of FIG. 7, the exhaust variable flow-restrictor 221 comprises one or more panels 221 which are pivotably mounted to the

generator **201**, for example to the chimney **202** or heating chamber **204**, such that pivoting the panels **221** causes the cross-sectional area of the flow exhaust **209** to be varied, although any other suitable means for varying the flow exhaust **209** cross-sectional area may instead be used.

(56) It is not necessary for the power generator **201** to require both the intake variable flow-restrictor **218** as well as the exhaust flow-restrictor **221** in order to control fluid flow through the generator **201**—either one, or both, may optionally be provided.

(57) The proximal end of the flow duct **203**, that is the end of the flow duct **203** which is opposite to the end fluidly coupled to the flow intake **205**, comprises a flow duct outlet **207** which is fluidly coupled to the heating chamber **204** such that fluid from the flow duct **203** may flow into the heating chamber **204**. In this embodiment, although it is not essential, the proximal end of the flow duct **203** is configured to arcuately flare radially outwards to form a generally or substantially conical shape. The flow duct outlet **207** comprises a circumferentially extending slot which comprises a plurality of radially spaced strakes **227** configured to segment the flow and to condition it in order to reduce the turbulence of the flow and may optionally be configured to provide laminar flow to the heating chamber **204**. The strakes **227** of this embodiment and all other embodiments is entirely optional and non-essential to the invention, but improves the efficiency of the generator **202**.

(58) The heating chamber **204** comprises an external wall **208** which is configured to transmit light radiation incident thereon such that fluid within the heating chamber **204** is heated by the transmitted light radiation. The heating chamber **204** therefore defines an internal fluid heating space **231** within which the fluid is heated by the light radiation transmitted by the external wall **208**. In this embodiment and in the others described herein, the light radiation may optionally be solar light radiation.

(59) In this embodiment, the external wall **208** is optionally made from plastic sheeting which is optionally supported by a support structure which may optionally comprise ribs such as longitudinally extending ribs **213** and/or laterally extending ribs **214**. Equally, glass or other such light transmitting materials may also be used for the external wall **208** instead of the plastic sheeting and which may not require a support structure.

(60) The external wall **208** by virtue of its arcuate shape is configured to substantially focus the light radiation incident on it into the internal fluid heating space **231** within the heating chamber **204**. Thereby, the power-generator **201** increases the heating effect of the transmitted light and improves the efficiency of the generator **201**. Optionally, to further improve the heating effect and efficiency of the generator **201**, the generator **201**, for example the external wall **208**, may comprise a lens element **211** configured to focus the light radiation transmitted by the external wall **208** within the heating chamber. The lens element **211** may have a focal point or focal locus **232** which is within the heating chamber but is spaced away from all interior surfaces thereof (e.g. spaced away from all light receiving surfaces thereof) such that the focused light radiation directly heats the fluid within the heating chamber, i.e. the fluid within the internal fluid heating space **231** of the heating chamber. Thus, the fluid is heated solely, or at least predominantly, directly by the transmitted light. Optionally, the lens element **211** may be configured to focus the transmitted light radiation to an area, locus or point **232** which is within the heating chamber **204**, specifically within the fluid heating space **231** but spaced away from all interior surfaces thereof (including the external wall **208**, for example all surfaces which define the fluid heating space **231**) such that the focused light radiation directly heats the fluid within the heating chamber **204**. In the embodiment of FIG. 7, the arcuate external wall **208** itself is formed of panels which each comprise a lens element **211**. Thus, the external wall **208** itself may be configured so as to focus the transmitted light radiation to an area, locus or point **232** which is within the heating chamber **208** but spaced away from all interior surfaces thereof such that the focused light radiation directly heats the fluid within the heating chamber **204**. The locus **232** may extend circumferentially around the flow duct **203** within the heating chamber **204** to form a heating zone which is substantially ring-shaped or

toroidal.

(61) As in some embodiments the arcuate shape of the external wall **208** itself may form the lens element **211**, the external wall **208** may optionally comprise two spaced-apart layers **214**, **216** of material. The two layers of material may have an air-gap therebetween to reduce the thermal losses to the environment or the two layers may have a space therebetween which is configured to be filled with water, or another suitable liquid, thereby improving the focusing effect of the lens element **211**.

(62) The flow duct **203** comprises a convergent divergent nozzle **217**. The convergent divergent nozzle **217** is configured to accelerate the fluid flow within the flow duct **203**. An axial turbine **210** is provided at the throat of the nozzle **217**, such that it is exposed to the flow within the flow duct **203**. Providing the turbine **210** at the throat of the nozzle **217** means that it is provided with accelerated fluid flow as the nozzle **217** accelerates the flow from a speed which may not otherwise be optimal to drive turbine **210**. The turbine **210** is operably coupled to an electrical generator **222** such that rotation of the turbine causes the electrical generator **222** to generate electrical power. A shaft forms the operable connection between the turbine **210** and the electrical generator **222**, although other suitable operable connections may instead be used.

(63) The operation of the third embodiment of the convection-driven generator is substantially the same as that of the first. As light radiation **28** falls on the external wall **208** of the heating chamber **204**, it is transmitted by the external wall **208** and focused by the lens element **211** into the interior fluid heating space **231** of the heating chamber **204** which is filled with fluid to be heated.

(64) The focused light may in some embodiments be focused to a point or a locus **232** which is spaced away from the interior surfaces of the heating chamber **204** such that the transmitted and focused light directly heats the fluid within the heating chamber **204**. Thus the lens element **211** is configured to form a heating zone which is spaced away from the interior walls or surfaces of the heating chamber **204** such that the transmitted and focused light directly heats the fluid within the heating chamber. Thus, the fluid within the heating chamber **204** in some embodiments may not be heated by conduction, or at least not predominately by conduction, e.g. by conduction of heat from an interior surface of the heating chamber **204**, but is instead heated at least predominately, and in some embodiments entirely, directly by the transmitted light within the heating zone.

(65) As the fluid within the heating chamber **204** is heated by the transmitted light radiation **28**, it begins to rise upwards by convection and escapes the heating chamber **204**, and the generator **201**, through the flow exhaust **209**. The temperature to which the fluid is heated may be controlled by the exhaust variable flow restrictor **221** by controlling the rate at which fluid is able to flow out of the heating chamber **204**, therefore how long it is within the fluid heating space **231** and therefore how long it is exposed to the transmitted light radiation **28**. Thus, the exhaust variable flow restrictor **221** is pivotable to a position in order to achieve the desired temperature within the heating chamber **204** and thereby the desired fluid flow rate out of the heating chamber **204**. Once the heated fluid passes through the exhaust variable flow restrictor **221**, it is then exhausted to the ambient surroundings.

(66) The escape of heated fluid from the heating chamber **204** causes a reduced pressure in the heating chamber **204** which causes more fluid to be drawn into the heating chamber **204** via flow duct outlet **207**, thereby pulling fluid down the flow duct **203** and establishing a fluid current through the flow duct **203**. This in turn causes un-heated, ambient air to enter the generator via the flow duct **203** via flow intake **205**, the flow rate of the un-heated fluid being controllable via the intake variable flow restrictor **218**.

(67) The established fluid current through the flow duct **203** drives rotation of the axial turbine **210** which thus generates electrical power via electrical generator **222** operably coupled to the turbine **210** via a shaft which connects the electrical generator **222** to the turbine **210**.

(68) With reference to FIGS. **8** and **9**, these figures show the axial turbine of FIG. **10** in top and bottom view respectively and as such show that the turbine has a plurality of turbine blades **25**

arranged circumferentially around the turbine. It is these turbine blades **25** which are exposed to the flow within the flow duct **3**. As shown in FIG. **10**, the turbine blades **25** may comprise a plurality of turbine blade stages, each producing increased flow pressure along the turbine **10**, but this is entirely optional. The turbine **10** of FIGS. **8** to **10** is not limited to use in the convection-drive power generator of the present invention but may be used in any scenario requiring a turbine **10** which is capable of operating within a wider range of flow speeds without stalling.

Claims

1. A convection-driven power generator comprising: a flow intake configured to supply fluid to the generator; a flow duct having a duct inlet and a duct outlet wherein the duct outlet is spaced downstream from the duct inlet along the flow duct, the duct inlet being fluidly coupled to the flow intake; a heating chamber fluidly coupled to the duct outlet so as to receive fluid from the duct outlet, the heating chamber comprising an external wall configured to transmit light radiation incident thereon such that fluid within the heating chamber is heated by the transmitted light radiation; a flow exhaust fluidly coupled to the heating chamber and configured to exhaust fluid heated by the heating chamber from the heating chamber; a turbine arranged within the flow duct, downstream of the flow intake, and exposed to fluid flow through the flow duct such that when fluid flows through the flow duct the turbine is caused to rotate by the fluid flow; and at least one lens element configured to focus the light radiation transmitted by the external wall within the heating chamber; wherein the generator is a light-electrical power generator configured to convert incident light radiation into electrical power, and wherein the heating chamber is an oblate toroid in shape, wherein an annulus of the toroid is arranged along the toroid axis of the heating chamber, and wherein the heating chamber extends circumferentially around the flow duct and the flow duct extends out of the heating chamber.
2. The convection-driven power generator of claim 1, wherein the convection-driven power generator is configured such that, when fluid is heated within the heating chamber by the transmitted light radiation, the heated fluid is caused to flow through the flow exhaust via convection and thereby establishes a fluid current within the flow duct which causes fluid to enter the duct inlet and in turn to enter the generator through the flow intake.
3. The convection-driven power generator of claim 1, wherein the lens element is configured to substantially focus the transmitted light radiation to an area, locus or point which is within the heating chamber but spaced away from all interior surfaces thereof such that the focused light radiation directly heats the fluid with the heating chamber.
4. The convection-driven power generator of claim 1, further comprising a chimney, the chimney comprising at least one of the flow duct or the flow intake.
5. The convection-driven power generator of claim 4, wherein the flow exhaust extends peripherally around the chimney or flow duct.
6. The convection-driven power generator of claim 1, wherein the heating chamber comprises an annulus extending therethrough, and wherein the flow duct, or a portion thereof, passes longitudinally through the annulus.
7. The convection-driven power generator of claim 1, wherein the external wall is made from plastic sheeting supported by a rib support structure.
8. The convection-driven power generator of claim 1, wherein the flow duct comprises a convergent-divergent nozzle for accelerating the fluid flow, the turbine being provided within the nozzle.
9. The convection-driven power generator of claim 1, wherein the flow intake comprises a variable flow restrictor configured to vary the fluid flow rate flowable through the generator.
10. The convection-driven power generator of claim 1, wherein the flow exhaust comprises a variable flow restrictor configured to vary the cross-sectional area of the flow exhaust so as to

control the rate of fluid flow flowable through the generator.

11. The convection-drive power generator of claim 10, wherein the flow exhaust variable flow restrictor is pivotably mounted to the generator.

12. The convection-driven power generator of claim 1, wherein the external wall of the heating chamber is configured to transmit solar radiation incident thereon such that fluid within the heating chamber is heated.

13. The convection-driven power generator of claim 1, wherein the turbine is operably connected to an electrical generator such that rotation of the turbine causes the electrical generator to generate electrical power.

14. The convection-driven power generator of claim 1, wherein the flow duct comprises stator blades arranged upstream of the turbine, the stator blades being configured to cause the flow to spiral down the flow duct before it passes through the turbine.

15. The convection-driven power generator of claim 1, wherein the turbine comprises a chamber containing a viscous liquid, wherein the chamber is configured such that the viscous liquid is caused to flow radially outwards within the chamber through centrifugal acceleration of the viscous liquid upon rotation of the turbine.

16. The convection-driven power generator of claim 1, a portion of the flow duct downstream of the turbine arcuately flares radially outwards to form a generally or substantially conical shape, the end of the flared portion comprising the duct outlet being fluidly coupled to the heating chamber.

17. The convection-driven power generator of claim 1, wherein the flow intake is configured to receive fluid from the external environment.

18. The convection-driven power generator of claim 1, wherein the power generator is generally or substantially spherical or spheroid.
